

Homework 1

Due Friday, May 15

1. List the five problems that DeFi could solve (they are listed in the introductory slides).
2. Define in a sentence or two a "ledger".
3. Describe in a sentence or two the difference between a "centralized ledger" and a "distributed ledger".
4. Is blockchain a centralized or distributed ledger?
5. To certify the identity of the user ordering a transaction, the user attaches a digital signature. Describe briefly how a digital signature is created.
6. A user that orders a transaction in the blockchain attaches a digital signature. Conceptually, this is similar to the signature we write on a paper check when we make a payment. If I want to verify the authenticity of the signature, do I need to know the "private key" of the user that ordered the transaction? Explain briefly.
7. A hashing algorithm converts any digital object in a new digital object of fixed length. Using the hash function SHA-256, compute the hash value for the following text:

"On September 5 2022 John transferred 0.1 bitcoin to Mary"

Now recompute the hash value for a slightly modified text:

"On September 5 2022 John transferred 0.1 Bitcoin to Mary"

The difference in the two sentences is that in the first "bitcoin" is written with small "b" while in the second is written with big "B". Copy and paste in your answer the computed hash values. Are they similar given that the text inputs are similar? To compute the hash value you can use the following link:

<https://emn178.github.io/online-tools/sha256.html>

8. Are all nodes in a Bitcoin blockchain miner? What is the task of miners in a bitcoin blockchain?
9. How are miners in a bitcoin blockchain compensated?
10. Briefly explain the differences between a validation system based on "proof-of-work" and a validation system based on "proof-of-stake".